7.1: Functions Defined on General Sets

Definition.
A **function** f **from a set** X **to a set** Y, denoted $f : X \to Y$, is a relation from X, the **domain** of f, to Y, the **co-domain** of f, that satisfies

1. every element in X is related to some element in Y, and
2. no element in X is related to more than one element in Y.

Then, given any element $x \in X$, there is a unique element in Y that is related to x by f:

$$\forall x \in X,\ f(x) \in Y$$

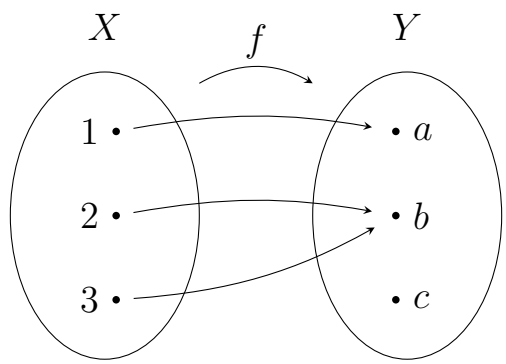
The set of all values of f taken together is called the **range of** f or the **image of** X **under** f:

$$\text{range of } f = \{y \in Y \mid y = f(x), \quad \text{for some } x \in X\}.$$

Given an element $y \in Y$, there may exist elements in X with y as their image. When x is an element such that $f(x) = y$, then x is called a **a preimage of** y or **an inverse image of** y:

$$\text{the inverse image of } y = \{x \in X \mid f(x) = y\}.$$

Example. Using the arrow diagram below, identify the following:



domain and co-domain

$\{1, 2, 3\}$ $\{a, b, c\}$

range

$\{a, b\}$

$f(1) = a$ $f(2) = b$ $f(3) = c$

Is 2 an inverse image of b? Is 2 an inverse image of c?

Yes No

Find the inverse image of a, b, and c

inverse image of $a = \{1\}$
inverse image of $b = \{2, 3\}$
inverse image of $c = \varnothing$

Represent f as a set of ordered pairs

$\{(1, a), (2, b), (3, b)\}$

Example. Given a set X, define a function $I_X : X \to X$ (**identity function on** X):

$$I_X(x) = x, \ \forall x \in X$$

What are the $\underbrace{\text{domain}}_{X}$, $\underbrace{\text{co-domain}}_{X}$, and $\underbrace{\text{range}}_{X}$ of I_X?

Definition.

For $b > 0$ and $b \neq 1$, the **logarithmic function**

$$y = \log_b(x) \qquad \text{(logarithmic form)}$$

has domain $x > 0$, base b, and is defined by

$$b^y = x \qquad \text{(exponential form)}$$

Example. Find the following:

$\log_3(9) = 2$

$3^x = 9$

$\log_2\left(\frac{1}{2}\right) = -1$

$2^x = \frac{1}{2}$

$\log_{10}(1) = 0$

$10^x = 1$

$\log_2\left(2^m\right) = m$

$2^x = 2^m$

$2^{\log_2(m)}, \ (m > 0)$

$= m$

Definition.
If $X \to Y$ is a function and $A \subseteq X$ and $C \subseteq Y$, then

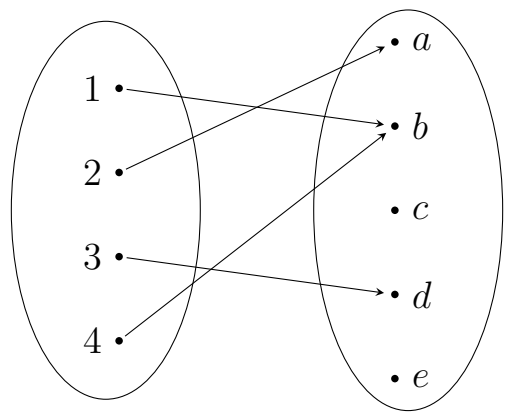
$$f(A) = \{y \in Y \mid y = f(x) \text{ for some } x \in A\}$$

and

$$f^{-1}(C) = \{x \in X \mid f(x) \in C\}.$$

$f(A)$ is called the **image of** A, and $f^{-1}(C)$ is called the **inverse image of** C.

Example. Let $X = \{1, 2, 3, 4\}$ and $Y = \{a, b, c, d, e\}$, and define $F : X \to Y$ by the following arrow diagram:



Let $A = \{1, 4\}$, $C = \{a, b\}$, and $D = \{c, e\}$. Find

$F(A) = \{b\}$

$F(X) = \{a, b, d\}$

$F^{-1}(C) = \{1, 2, 4\}$

$F^{-1}(D) = \emptyset$